CURRICULUM VITAE

# Darah N. Oxford

PhD Candidate · University of Missouri oxforddn@missouri.edu Columbia, MO

ABOUT

## Profile

A versatile scientist and educator whose background spans animal communication, neuroethology, physiology, ecology, entomology, and nursing science. Brings PhD-level research in neuroethology, hands-on clinical nursing experience across ICU, cardiac, and rehabilitation settings, and a demonstrated record of innovative course design. Deeply invested in active learning, undergraduate research mentorship, and the practical integration of AI tools into evidence-based pedagogy.

Intro & General Biology · Human Physiology & Anatomy · Neuroscience & Animal Behavior · Ecology & Entomology · Research Methods & Biostatistics · Nursing & Health Sciences · AI Integration in Higher Ed · Science of Learning · Plant Biology & Pollinator Ecology

CREDENTIALS

## Education

### PhD, Biological Sciences
University of Missouri – Columbia

Expected May 2027 · Adviser: Dr. Johannes Schul
Dissertation: Neural circuits underlying call pattern generation in *Neoconocephalus* katydids
Pillars of Discovery Writing Fellow, 2026

### BS, Nursing
Southern Arkansas University

May 2018 · Summa Cum Laude · GPA 3.92
Sigma Theta Tau · President's List: Fall 2016, Spring 2017, Fall 2017

### MS, Entomology
University of Missouri

### BS, Biology
University of Missouri

May 2005 · Cum Laude · GPA 3.75

December 2010 · GPA 3.6
Native plant characteristics & insect load

EXPERIENCE

# Professional Activity

### Guest Lecturer — Integrative Psychology (PSYC 495)

Spring 2024 & 2026

Columbia College, MO

- Statistical Design Methods for senior capstone; invited to return based on student and faculty feedback.

### Course Developer & Lab Instructor — Human Physiology (Bio 3700)

2023–Present

University of Missouri

- Designed new laboratory from scratch: lesson plans, instructor resources, Canvas LMS site for 200+ upper-level pre-health students per semester. Successfully implemented Fall 2023; continues every semester.
- Fall 2026: teaching two lab sections (24 students each) of upper-level pre-professional students; integrates clinical nursing experience (ICU, cardiac cath/EP lab, rehabilitation) into physiology instruction.

### Graduate Teaching Assistant — Bio 1500

Aug 2020–May 2023

University of Missouri

- Lab & discussion sections: team-based learning, health equity, scientific reasoning, data interpretation.
- Teaching effectiveness scores 4.72–4.80 / 5.0; 76th–89th percentile among all Division of Biological Sciences graduate TAs.

### Camp Co-Director & Science Lead — STEAM Summer Camp

July 2023

Columbia College, MO

- Week-long science camp for grades 3–8; led the science program: plant biology, insect anatomy, microscopy, chromatography, UV light.
- Built a community–university partnership of 15 local organizations and businesses: sponsors such as 3M and Veterans United, and guest presenters including the Missouri Department of Conservation, the Columbia Aeronautics and Space Association (CASA) student aerospace and robotics club, Columbia College nursing simulation, and a local beekeeper.

### Mini-Course Instructor — Columbia STEM Alliance Summer Expedition

July 2021

Columbia, MO

- "Pretty Plants & Yucky Bugs" — co-taught with Dr. Jon Oxford; angiosperm anatomy, plant-pollinator interactions, entomology.

### Biology Lab Teacher — BIOME Intensive Orientation Experience

August 2022

University of Missouri

- Lab instruction for incoming first-year students.

### Peer Tutor — Nursing Program

2017–2018

Southern Arkansas University

- Individualized and group tutoring; nursing skills labs; pharmacology calculations.

### Supplemental Instructor — World Literature II

2016–2018

Southern Arkansas University

- Weekly discussion and exam prep sessions. Supplemental Instructor of the Year (2017); SI of the Month (Oct 2016, Feb 2017).

### Substitute Teacher

2011

Columbia Public Schools

- Extended full-time assignments; elementary, secondary, special education.

### ESL / English Conversation Instructor

2005–2007; 2012–2014

International Church, MU Campus & Community Center, Germany

- English conversation classes for international students and German community members.

## RESEARCH

### Doctoral Research — Neuroethology

2020–Present

University of Missouri

- Identifying neural circuits underlying call generation in *Neoconocephalus* katydids via neuroanatomy, electrophysiology, and behavioral ecology.
- Supervised two undergraduate research contributions: CNS circuit identification poster (J. Kelly-Cuenca, 2023) and wing coordination poster (B. Clemons, 2023).

### Master's Research — Entomology & Plant Sciences

2005–2010

University of Missouri

- Native plant characteristics and insect load; plant-pollinator interactions; experimental field and lab design.

### Hormone Assay Laboratory

June–Sept 2010

University of Missouri

- Enzyme-linked immunoassay: salivary testosterone and cortisol.

## CLINICAL — REGISTERED NURSE (MISSOURI LICENSED)

### Registered Nurse — Rusk Rehabilitation Hospital

Mar–Jul 2020

Columbia, MO

- Stroke, TBI, hip fracture, spinal cord, and amputee rehabilitation. Individualized care plans; interdisciplinary team; patient and caregiver education.

### Registered Nurse — Lutheran Senior Services REACH Rehabilitation

Jan–Aug 2020

Columbia, MO

- Person-centered care; post-acute rehabilitation unit.

### Registered Nurse — Cardiac Catheterization / Electrophysiology Lab

Aug–Dec 2019

University Hospital, Columbia, MO

- EP procedures: ablations, cardioversion, pacemaker/ICD implants, electrophysiology studies.

### Registered Nurse — ICU

Jun 2018–Jun 2019

Magnolia Regional Medical Center, Magnolia, AR

- Total ICU care, telemetry, outpatient procedures; floated to Emergency Room, Med-Surg, Surgery.

## AI & SCHOLARSHIP OF TEACHING

### Workshop Leader — Virtual Conference on Teaching (VCT)

Feb 2026

Society for the Teaching of Psychology (APA Div. 2)

- "Unlocking AI's Potential in Higher Ed" — 2-hour interactive workshop with live demos: agentic browser detection, split-screen assessment systems, discipline-specific course chatbots, interactive learning modules.

### Co-Creator & Curator — Imagination is the Key ↗

2025–Present

Public AI-in-Teaching Database (with Dr. Jon Oxford, Columbia College)

- Publicly searchable Notion database documenting 200+ real AI-in-teaching experiments; organized by topic, difficulty, and ROI. Anchors faculty workshops and national open office hours.

SCHOLARSHIP

## Publications & Presentations

Oxford, D. N., & Oxford, J. (In Preparation). Imagination Is the Key: AI experiments as faculty development. *Society for the Teaching of Psychology, Teaching Tips*.

Oxford, J., & Oxford, D. N. (In Preparation). Blueprint First: Building an AI-assisted assessment ecosystem. *Society for the Teaching of Psychology, Teaching Tips*.

Oxford, J., & Oxford, D. N. (2026). Imagination is the Key: A public database of daily AI experiments in teaching. *Society for the Teaching of Psychology, Teaching Tips*.

Oxford, D. N., & Oxford, J. (2026, October). Imagination Is the Key: AI experiments as faculty development [Poster]. Annual Conference on Teaching (ACT), Society for the Teaching of Psychology, St. Louis, MO. *Interactive poster on learning from public, documented AI experiments: entry points by teaching problem; how the trials grew from custom chatbots (Fall 2025) to building interactive demos and textbooks (Spring 2026) to AI agents; and one experiment, an AI browser taking a course quiz, followed from first notice to a redesigned assessment, with every wrong turn kept public.*

Oxford, J., & Oxford, D. N. (2026, October). Blueprint First: Building an AI-assisted assessment ecosystem [Symposium]. Annual Conference on Teaching (ACT), Society for the Teaching of Psychology, St. Louis, MO.

Oxford, J., & Oxford, D. N. (2026). Imagination is the Key: A public database model for faculty AI development [Poster, accepted; not presented]. AAC&U Conference on AI and Higher Education.

Oxford, J., & Oxford, D. N. (2026). Hamilton-Zuk Hypothesis. In Shackelford, T. K. (Ed.), *Encyclopedia of Sexual Psychology and Behavior*. Cham: Springer International Publishing.

Oxford, J., & Oxford, D. N. (2022). Testosterone: Digit ratios. In Shackelford, T. K. (Ed.), *Encyclopedia of Sexual Psychology and Behavior*. Springer.

Oxford, J., & Oxford, D. N. (2022). Testosterone: Paternal investment. In Shackelford, T. K. (Ed.), *Encyclopedia of Sexual Psychology and Behavior*. Springer.

Oxford, J., & **Oxford, D. N.** (2017). Co-sleeping. In Shackelford, T. K., & Weekes-Shackelford, V. A. (Eds.), *Encyclopedia of Evolutionary Psychological Science*. Cham: Springer. *(Coauthored as a B.S.N. student.)*

Schul, J., Murphey, K., **Oxford, D.**, Kelly Cuenca, J., & Favela, E. (2021). How does the central nervous system change when new call types evolve? Virtual Meeting of the Animal Behavior Society.

**Oxford, D. N.** (2018). Kangaroo care provides multiple benefits to preterm low birth weight infants. UAMS Nursing Research Conference. *Winner: Best BSN Poster & Podium Presentation.*

RECOGNITION

## Awards & Honors

- Featured in *The Buzz*, Division of Biological Sciences digital magazine — Schul lab research feature highlighting dissertation work (July 2026)
- Pillars of Discovery Summer Dissertation Writing Fellowship — Graduate School, MU (2026)
- Sandra K. Abell Science Education Award — Outstanding commitment to college science teaching (2024)
- Griffiths Leadership Society for Women — Inducted Fall 2020
- Dean's List, UAMS College of Nursing Nurse Practitioner Program (Spring 2019)
- Best Student Research Poster & Podium Presentation, UAMS Nursing Research Conference (2018)
- Supplemental Instructor of the Year Award & Scholarship (2017)
- Sigma Theta Tau, Nursing Honor Society — Inducted October 2017
- Southern Arkansas University President's List (4.0 GPA): Fall 2016, Spring 2017, Fall 2017

SERVICE & MEMBERSHIP

## Professional Memberships & Service

- Open Office Hours with the Oxfords: co-facilitator, drop-in AI support for faculty (Spring 2026–Present)
- Graduate Coalition Alliance (Spring 2022–Present)
- Sigma Theta Tau, Nursing Honor Society (2017–Present)
- Vice President of Nursing (Class of 2018), SAU (2016–2018)

- Biological Graduate Student Association (2020–Present)
- Coalition of Graduate Workers (Fall 2020–Present)
- Griffiths Leadership Society for Women (2010, 2020–Present)
- Arkansas Student Nurses Association, Projects Manager (2017–2018)
- C.V. Riley Entomological Society, Public Relations Officer (2009)

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